

of Aerofarms quipped that, they are always in capital-raising mode due to the nature of the business. When planning the finances of a vertical farming startup, you need to remember that even business activities like planning and hiring need money.

4. Funding experts suggest that if you wish to go into the business of vertical farming or vertical farming technology, you would have to start small and set a track record, while gradually increasing the size and complexity of projects. That way, your funding model will also be more manageable, as your growth can be partly organic and partly funded.

Second, you need to remember that most investors do not fund farms; they fund farming mechanisms and infrastructure!

5. Being an emerging field and one that the government is interested in promoting, you can expect funding from industry bodies like Confederation of Indian Industry, too. Some of the organic farming startups in the North East, for example, are funded by Confederation of Indian Industry-Ahmedabad and organisations like Village Capital, USA.

The wait before the cash register chimes

"I think commercial viability of vertical farming, its reliability, potential to scale to meet demand and the like are yet to be proven. Not to mention proof-of-value of soil-less agriculture from nutrition and health perspectives. Consequently, like any other revolution, we need both an abundance of timely resources and a good measure of restraint. We also need a concerted effort from several stakeholders, especially the government—not just subsidies, but rather complete ownership of infrastructure and policy pieces with respect to space, power, regulation,

Tech recipe for vertical farming

Grow lighting. Most commercial vertical farms use LED lights to provide plants with exactly the spectrum, intensity and frequency they need for photosynthesis in the most energy-efficient way possible. Philips is a key player in grow lighting.

Conditioning the air. Smart and energy-efficient large-area HVAC systems make it possible to grow veggies anywhere in the world, by maintaining humidity and temperature within a range that is optimal for photosynthesis.

Sensors, data and intelligence. Sensor networks are used to monitor and replenish nutrient levels in water or air. Other environmental aspects are also constantly monitored and studied remotely using predictive analysis to control the systems and to minimise the risks associated with traditional agriculture. For example, data scientists from Harvard University, USA, and Massachusetts Institute of Technology, USA, analyse data collected at Aerofarms using machine-learning software, to optimise the growing algorithms of 250 different types of plants. These growing algorithms can be replicated to ensure constant yield every time.

policy and tax breaks. This is about creating the ecosystem, not just funding pieces of the puzzle. The business model, ideally, should be a private and public partnership, for it to work in the long term," says Malhotra.

The good news is that the government is in favour of vertical farming. Last year's Vertical Farming Conference in Bengaluru was a step towards getting hi-tech agriculture within the reach of local farmers. Speaking to *The Hindu* at the conference, N.K. Krishna Kumar, deputy director general of ICAR, mentioned that the scope of vertical farming goes far beyond terrace gardening. It involves cultivation of various horticultural crops—fruits and vegetables, besides rare varieties of flowers such as orchids in multiple layers—with the help of proper supporting structures. He said it is possible to grow enough vegetables and fruits or flowers even in apartments through vertical gardening. Urban residents can grow their own food, or even take up commercial farming.

Irrespective of the business model, Malhotra has a word of caution for technopreneurs. "Technology players will fail regardless of the amount and term of funding if they continue to be siloed in their approach, as has been endemic in the sector. Technology companies,

including information technology, try to avoid solving the larger business problem and are reluctant to go beyond the scope of their own narrow expertise.

"Additionally, technology can help in forecasting demand and understanding customers' real needs. Focus has to be more on the consumer. Put simply, any technology needed to support the vertical farming business must be oriented to the customer (farmer), and prototype solutions should be quickly tested for efficacy.

"A design-thinking intervention, done seriously at the beginning of each exercise, will ensure a holistic view to bring out a coherent and sustainable solution, tested beforehand for efficacy and need fulfilment."

Vertical farming is fresh pasture for technopreneurs in India. Its benefits have, however, been proven abroad, especially in land-scarce regions like Singapore. With the right team and tech in place, it is possible to enter this segment profitably in India, too. But as Malhotra says, the key would be to set the business up in partnership with the right government bodies, to lower the risk and improve market reach. Even more important is the technologist's mindset—the ability to put aside technology and think like a farmer. **EFY**